

HealthConnect Data Analytics Deliverable

AnalystLab Africa Experience Lab | Data Analytics Track | Tool: Power BI

1. Project Overview

HealthConnect business problem: HealthConnect Clinic, a fictional healthcare provider, is experiencing missed appointments, inefficient use of appointment slots, and repetitive patient enquiries. The clinic wants to use data and AI to improve decision-making and the patient experience.

Central project question: How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

Data Analytics role: This track is responsible for understanding the appointment dataset, identifying the variables most associated with attendance and no-shows, defining the business questions the data can answer, and proposing KPIs and an analysis approach that will guide deeper analysis and dashboard-building (Power BI) in later weeks.

2. Dataset Overview

Purpose: The HealthConnect_Appointment_Data.csv file contains fictional, anonymised appointment-level records used to study attendance behaviour and support a no-show reduction strategy.

Size: 5,000 records across 18 variables, spanning appointment dates from January 2025 to June 2026.

Main categories of variables:

- Patient demographics — gender, age, age_group
- Appointment details — appointment_type, booking_date, appointment_date, appointment_day, appointment_time
- Booking behaviour — booking_lead_days, previous_appointments, previous_no_shows
- Reminders — reminder_sent, reminder_channel
- Logistics — distance_to_clinic_km, waiting_time_minutes
- Outcome — appointment_outcome

Appointment outcome breakdown:

Outcome	Count	% of Total
No-Show	2,423	48.5%
Attended	2,314	46.3%
Cancelled	263	5.3%

No-shows are the single largest outcome category — larger than attended appointments — which confirms this is a meaningful operational problem worth investigating.

3. Data Dictionary Review

Important variables and what they contribute to the analysis:

Variable	Contribution to analysis
age / age_group	Tests whether certain age bands are more prone to missing appointments
appointment_type	Checks whether no-shows differ by type of visit (e.g. Follow-up vs Diagnostic Test)

Variable	Contribution to analysis
booking_lead_days	Tests whether appointments booked far in advance are more likely to be missed
previous_appointments / previous_no_shows	Establishes attendance history as a predictor of future no-shows
reminder_sent / reminder_channel	Assesses whether reminders — and which channel — relate to attendance
distance_to_clinic_km	Tests whether patients further from the clinic are more likely to miss appointments
waiting_time_minutes	Tests whether longer expected waits relate to no-shows or cancellations
appointment_outcome	The outcome variable that all other variables are analysed against

All 18 variables in the dataset were reviewed against the Data Dictionary; no undocumented or unexpected variables were found, and recorded example values (e.g. gender categories, appointment_time bands) matched the dictionary's descriptions.

4. Initial Data Quality Assessment

Check	Finding
Missing values	reminder_channel: 1,366 missing; distance_to_clinic_km: 90 missing; waiting_time_minutes: 60 missing
Duplicate records	None — appointment_id is unique across all 5,000 rows, and no fully duplicate rows exist
Data types	Consistent with the Data Dictionary (text, integer, decimal, date fields all typed as expected)
Invalid / inconsistent values	None found — previous_no_shows never exceeds previous_appointments; booking_date is never after appointment_date
Potential outliers	None flagged — age (18–80), booking_lead_days (0–60), distance (0.5–45km) and waiting_time (2–68 min) all fall within plausible clinical ranges
Target/outcome consistency	appointment_outcome contains only the three documented categories (Attended, No-Show, Cancelled) with no typos or stray values

Note on missing reminder_channel: the 1,366 missing values align exactly with the 1,366 rows where reminder_sent = "No" — the channel is blank because no reminder was sent, which is expected structure rather than a data quality defect. The missingness in distance_to_clinic_km and waiting_time_minutes is low in volume (under 2% of records) and appears to be genuine, unexplained missing data that will need a documented handling decision (e.g. imputation or exclusion) before KPI calculation in Week 5.

5. Key Variables for No-Show Analysis

- Patient demographics — gender, age, age_group
- Appointment/booking information — appointment_type, booking_lead_days, appointment_day, appointment_time
- Previous attendance history — previous_appointments, previous_no_shows
- Reminder information — reminder_sent, reminder_channel
- Distance — distance_to_clinic_km
- Waiting time — waiting_time_minutes

- Appointment outcome — appointment_outcome (target variable)

An early, purely descriptive look supports treating attendance history and reminders as relevant: the no-show rate rises from 44% for patients with 0 previous no-shows to 68% for patients with 3 previous no-shows, and the no-show rate is somewhat lower when a reminder was sent (47.4%) than when it wasn't (51.4%). These are preliminary observations from Week 4 exploration, not final findings — full segmentation and relationship analysis is planned for Week 5.

6. Business Questions

- What proportion of appointments are missed?
- Which patient or appointment characteristics are associated with no-shows?
- Does reminder status (and channel) relate to attendance?
- Does waiting time influence attendance?
- Does distance from the clinic relate to missed appointments?
- How does previous no-show history relate to future attendance?

7. Potential KPIs

KPI	Business Question	Why It Matters
Appointment No-Show Rate	How frequently are appointments missed?	Measures the scale of the problem
Attendance Rate	What proportion of appointments are attended?	Tracks overall appointment utilisation
Reminder Response / Attendance Rate	Are reminders associated with better attendance?	Helps assess reminder effectiveness
Repeat No-Show Rate	Are patients with previous no-shows more likely to miss again?	Helps identify higher-risk appointment groups
Appointment Slot Utilisation Rate	How efficiently are available slots being used?	Connects attendance to operational efficiency

Per the Week 4 scope, these KPIs are identified and justified only — calculation and visualisation are planned for later weeks using Power BI.

8. Initial Analysis Approach

- Data inspection — confirm structure, types, and value ranges against the Data Dictionary (completed this week)
- Data cleaning — decide and document a handling approach for missing distance_to_clinic_km and waiting_time_minutes values before loading into Power BI
- Descriptive analysis — summarise appointment volumes by type, day, time, and demographic group
- No-show segmentation — break down no-show rate by each key variable identified in Section 5
- Relationship analysis — examine how reminder status, distance, waiting time, and prior no-show history relate to the outcome
- KPI development — calculate the KPIs defined in Section 7 in Power BI
- Visualisation — build a Power BI dashboard presenting KPIs and segment breakdowns
- Business recommendations — translate findings into practical suggestions for reducing no-shows
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9. Assumptions, Limitations, Risks & Dependencies

- Assumption: the dataset is a representative, fictional simulation of real clinic behaviour, so patterns found are illustrative rather than clinically validated
- Limitation: distance_to_clinic_km and waiting_time_minutes each have a small number of missing values (90 and 60 records respectively) that must be handled before KPI calculation
- Limitation: the dataset provides no information on appointment cost, staff availability, or clinic capacity, which may also affect no-shows but cannot be analysed here
- Risk: KPIs and segment findings from Week 4/5 are descriptive only — care is needed not to imply causation (e.g. reminders "causing" better attendance) without further testing
- Dependency: Week 5 KPI calculation and dashboard-building depend on finalising the missing-data handling approach agreed in this document
- Ethical consideration: patient data is anonymised and fictional, but any real-world extension of this analysis would need to follow the clinic's data protection obligations